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Theoretical and experimental study of time domain-induced polarization in water-saturated sands

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Abstract

A theoretical model of spectral-induced polarization (IP) of sand is presented. In the proposed model, contacts of sand grains and intergrain solution-filled space are considered as electrical current passages of varying thickness, which differ in values of ion transport number. Ion-selective narrow passages are considered as active zones, large passages as passive. The proposed model describes spectral IP characteristics for the medium where the length of passive zones is much greater than the length of active ones. The model is called short narrow pores (SNP) model. The SNP model predicts a growth of IP time constant with increase of length of ion-selective zone. Both the time domain and frequency domain parameters are described. The parameters of Cole–Cole model corresponding to the SNP model were also found. The behaviour of model parameters is compared with experimental data obtained on natural and sieved sands using time domain technique. The natural sand spectra correspond neither to the simple SNP model nor simple Cole–Cole model with single time constant because the lengths of ion-selective zones vary, reflecting the grain-size distribution. The spectra of sieved sand compared with the theoretical SNP spectra reveal close correspondence between experimental data and theoretical parameters. For four sieved sands, both the theoretical and experimental data show that the time constant of the IP is proportional to the square of the average grain size.
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1. Introduction

The problem of distinguishing between sand units containing saline water and clay units on the basis of surface electrical methods is very important in environmental investigations at oil production fields. The DC or low frequency resistivity of these units is frequently almost the same. Thus, the application of induced polarisation (IP) is recommended. We started this work with the purpose to use IP spectral characteristics to distinguish between different types of rocks and soils.

The phenomenon of induced polarization in rocks has been discovered by Schlumberger (1920). Pelton et al. (1978), Komarov (1980), Olhoeft (1985) and others studied the characteristics of spectral-induced polarization (SIP) in frequency domain (FD) or time domain (TD). Most of these studies were conducted in

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the systems that included both electron and ion conductors (ore minerals and pore moisture). The most common approach is an approximation of frequency dispersion in rocks (both electron-ion and ion-conductive) using the Cole–Cole equation (Pelton et al., 1978):

$$\rho(\omega) = \rho_0\left[1 - \eta_0\left(1 - \frac{1}{1 + (i\omega\tau)^c}\right)\right], \quad (1)$$

where ρ is the complex resistivity (in Ω·m), ω is the angular frequency of the polarizing current (in s^{-1}), ρ_0 is the DC resistivity (in Ω·m), $i=(-1)^{1/2}$, $\eta_0 = U_p/U_i$ is the chargeability (in V/V) corresponding to polarization seen immediately after switch-off of an infinite long current step (U_p and U_i are the electrical potential drops in off-time and on-time, respectively), τ is the time constant (in s) and c is the dimensionless exponent.

The time domain analogue of the Cole–Cole equation (Pelton et al., 1978) describes the transient IP characteristic during switching-on of the polarizing current (on-time):

$$\eta(t) = \eta_0\Psi_c(t) = \eta_0\left[\sum_{k=1}^{\infty}(-1)^{k-1}\frac{(x)^{kc}}{\Gamma(1+kc)}\right] \quad (2)$$

where $\Psi_c(t)$ is the normalised Cole–Cole transient characteristic that vary between 0 and 1, $\Gamma(z)$ is the gamma-function, $x = t/\tau$ is the dimensionless argument and t is on-time counted from the beginning of the current step.

The empirically defined constants (η_0, τ and c) depend on the petrophysical properties of rocks and soils. However, it is not entirely clear which coefficients correspond to which properties of the rocks.

Marshall and Madden (1959) offered a theoretical model of IP (formulated in terms of frequency domain) based on the membrane polarization. In the model, sediments are liken to serial connection of active (ion-selective) and passive (nonselective) zones. The zones have different numbers of cation and anion transport, which produces local concentration gradients under applied external voltage. The local solute flows, appearing under these concentration gradients, result in additional electrical current in ion-selective zones, which yields additional electrical field and causes the frequency dependency of the resistivity. The model predicts increase of the time constant with the growth of nonselective zones.

Both the Cole–Cole and Marshall–Madden models predict a maximum in the phase spectrum.

Fridrikhsberg and Sidorova (1961) developed similar approach (in terms of time domain) to predict chargeability of ion-conductive rocks. They expressed chargeability as follows:

$$\eta_0 = \frac{4(\Delta n)^2}{\left(\frac{l_1}{S_1} + \frac{l_2}{\alpha_2 S_2}\right)\left(\frac{S_1}{l_1} + \frac{S_2}{l_2}\right)} \quad (3)$$

where Δn is the difference of ion transport numbers in the active and passive zones; l_1, S_1, l_2, S_2 are the lengths and sections of passive and active zones, respectively; α_2 is the coefficient of efficiency of active zone, which depends on the surface conductivity and represents a ratio of the average capillary conductivity and the bulk solution conductivity.

Based on this equation, Fridriksberg and Sidorova estimated the chargeability of water saturated soils to be 1…2%. The artificial model containing serial connection of selective membranes (cellophane or collodion) and passive zones (capron cloth) was experimentally studied. The predicted and experimental magnitudes of chargeability were found to be close, which supports accuracy of the model.

Kormiltsev (1963) offered the TD formulation of Marshall–Madden model and performed calculations showing influence of current waveform on the measured decay. The lengths of active and passive zones were assumed to be equal. The resulting formula was derived from basic solution of the equation of diffusion, and represented infinite series. The IP decay was found to be dependent on the difference in ion transport number in active and passive zones, the diffusion coefficient and the length of the zones.

Numerous empirical data on SIP in soils were published for the low frequency range (10^{-2}…10^4 Hz) (Vinegar and Waxman, 1984; Börner et al., 1993, 1996; Vanhala, 1997, etc.). Three essential types of spectra emerge from these data.

(1) The phase spectrum is nearly constant over the observed frequency range. Therefore, the IP spectrum corresponds to neither Cole–Cole nor Marshall–Madden models. This type of spectra produce the Constant Phase Angle model (CPA) (Vinegar and Waxman, 1984; Börner et al., 1993, 1996).

(2) The phase spectrum has a maximum. Therefore, IP can be approximated by the Cole–Cole equation (Eq. (1)). The Marshall–Madden model can be also applied. Examples of such spectra are published in Vanhala, 1997.

(3) The phase spectrum increases with frequency increase (Vanhala, 1997). This type of spectra corresponds to neither the Cole–Cole nor the CPA models.

The goal of this paper is further development of spectral IP theory for water-saturated sand, based on

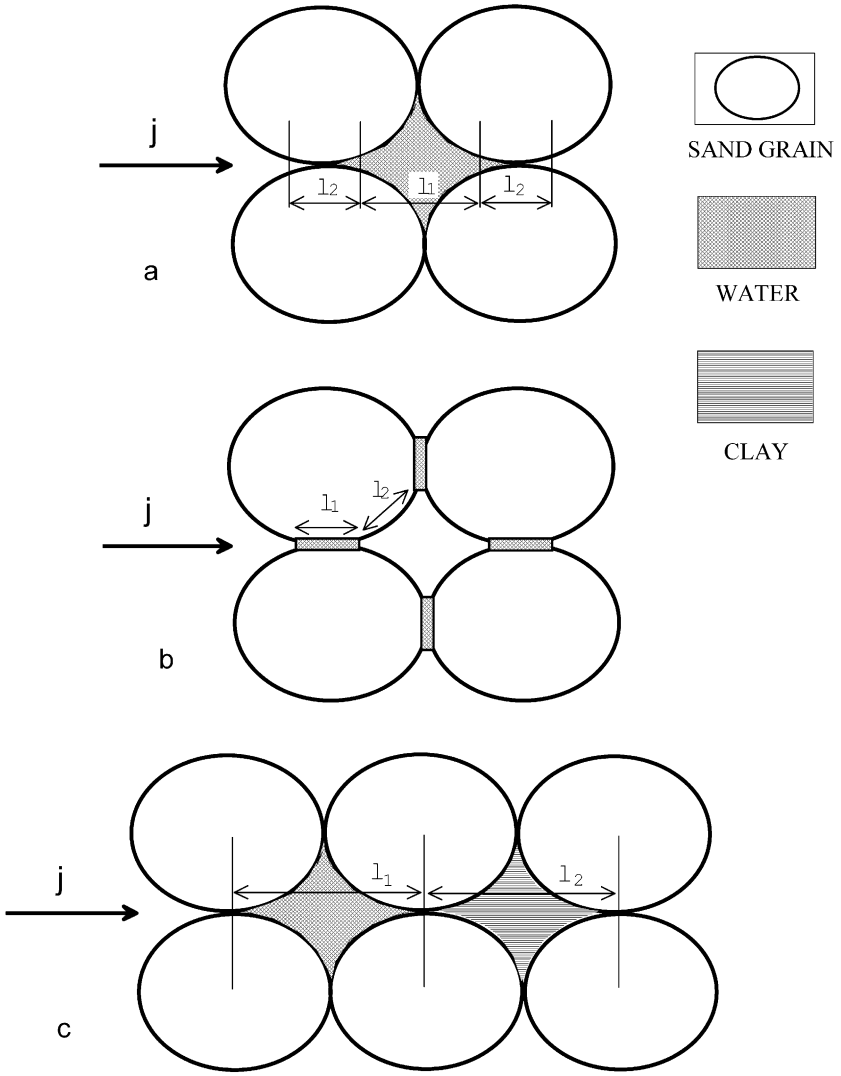


Fig. 1. Schematization of the electrical current passageway geometry in sandy-clay sediments. (a) Pure water-saturated sand. (b) Pure undersaturated sand. (c) Saturated sandy-clay sediments. In the saturated sand, water-filled intergrain space (where physico-chemical characteristics of the water are close to those of a free solution) can be considered as large pores of length l_1. Areas of grain contact, where the water is in a closer bond on the grain surface, can be considered as narrow pores of length l_2. In the undersaturated sand, the water film at the surface of moistened grains can be considered as a relatively narrow passage for electrical current (narrow capillaries of length l_2). Areas of grain contact, where water is more abundant, can be viewed as large capillaries of length l_1. In sandy-clay water-saturated sediments, the large passages of the electrical current are water-filled pores between the sand grains (l_1), while the narrow pores are those filled with clay particles (l_2).

the fundamental Marshall–Madden model, as well as experimental verification of this theory. Because experiments were conducted in TD, theoretical assessments were also formulated in TD. The correspondence between TD and FD characteristics is described in Appendix C.

2. Theory

In ion-conductive rocks and soils, the electrical current passes through interconnected serial and/or parallel large and narrow pores. The scheme showing these large and narrow current passages in soils (Fig. 1) does not cover all possible types of pore space. However, it appears to be sufficient for our IP modelling. Let us assume that the pore moisture contains one type of cations and one type of anions, both of valence 1. The IP phenomenon is caused by the difference between the ion transport numbers in large and narrow pores (Marshall and Madden, 1959; Fridrikhsberg and Sidorova, 1961; Vinegar and Waxman, 1984). This difference, in turn, is stipulated by different concentrations of cations and anions in the diffusion zone of the electrical double layer (EDL) (Fig. 2a) (Kruyt, 1952). Since the surface of the solid phase of soils is usually charged negatively, the cation concentration in the pore solution is higher than that of the anions. The thinner the pores, the greater the part of the pore volume occupied by EDL. Therefore, the greater the difference in average concentration between the cations and anions (Kruyt, 1952). It was shown that the electrical current flow through a sequence of large and narrow capillaries (Fig. 2b) results in deficiency of anions and surplus of cations at the boundaries of the thick and thin capillaries (Marshall and Madden, 1959; Fridrikhsberg and Sidorova, 1961). The cations surplus/deficiency q at

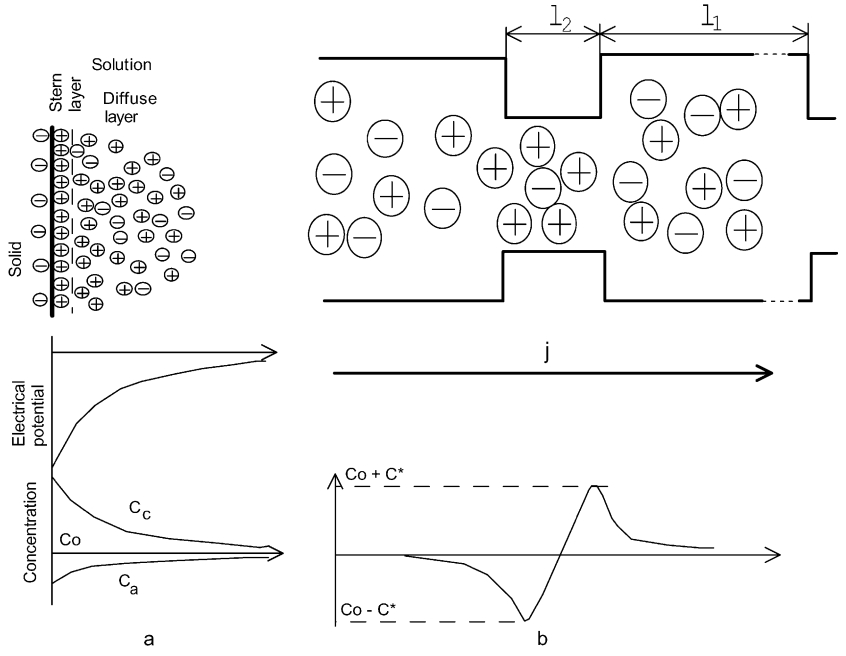


Fig. 2. IP scheme in ion-conductive rocks (according to Fridrikhsberg and Sidorova, 1961). (a) Electrical double layer (EDL) scheme, distribution of equilibrium potential and concentration in EDL. (b) Pore system and distribution of concentration under the influence of electrical current. For each pore, l_1, S_1, l_2, S_2 are lengths and sections. Narrow pore is characterised also by the coefficient of efficiency α_2 (see also Section 1). C_0—equilibrium concentration. C_c, C_a—cation and anion concentrations, respectively. C^*—additional concentration under the influence of electrical current.

the large/narrow capillary boundary can be described as follows:

$$q = \pm \frac{\mathrm{d}m}{\mathrm{d}t} = \frac{\Delta n j}{zF} \quad (\text{in mol s}^{-1}) \tag{4}$$

where j is the electrical current that flows through the capillary (in A), $z=1$ is the valence of the ions, F is Faraday's constant (in C/mol), $\Delta n = n_2{}^{\mathrm{c}} - n_1{}^{\mathrm{c}} = n_1{}^{\mathrm{a}} - n_2{}^{\mathrm{a}}$ is the difference of ion transport numbers in the large and narrow capillaries (n—ion transport number; the lower index 1—in large capillaries, the lower index 2—in narrow capillaries, the upper index c—cations, the upper index a—anions).

For small current density, we consider ion transport numbers $n(t)$ as invariable with time because the variation in the concentration is much smaller than the equilibrium concentration (the linear IP).

The total density of the pulse electrical current J^{t} includes a migration (Ω) component J^{Ω} and a diffusion (IP) component J^{d}, which can be expressed using the Ohm's and Fick's laws as:

$$\begin{aligned} J^{\mathrm{t}}(\boldsymbol{r},\ t) &= J^{\Omega}(\boldsymbol{r},\ t) + J^{\mathrm{d}}(\boldsymbol{r},\ t) \\ &= \sigma E^{\Omega}(\boldsymbol{r},\ t) - FRTu(\text{grad } C_{\mathrm{c}}(\boldsymbol{r},\ t) \\ &\quad - \text{grad } C_{\mathrm{a}}(\boldsymbol{r},\ t)) \end{aligned} \tag{5}$$

where σ is the conductivity of the solution (in $\Omega\cdot\mathrm{m})^{-1}$, E^{Ω} is the migration component of electrical field (in V/m), R is the universal gas constant (in $\mathrm{J}\cdot(\mathrm{mol}\cdot\mathrm{K})^{-1}$), T is the absolute temperature (in K), u is the ion mobility in $(\mathrm{m}\cdot\mathrm{mol}\cdot\mathrm{N}^{-1}\mathrm{s}^{-1})$, $C_{\mathrm{c}}(\boldsymbol{r}, \mathrm{t})$ and $C_{\mathrm{a}}(\boldsymbol{r}, \mathrm{t})$ are the cation and anion average concentrations in the pore solution (in mol/m^3), $\boldsymbol{r}$ is the vector of space coordinates and t is the time.

Note that the diffusion coefficient and ion mobility are related as: $D = RTu$.

The ion average concentrations in pores C_{c} and C_{a} are linearly related to the equilibrium concentration in bulk solution C (that corresponds to concentration on the pore axis for the pore much larger than the Debye length):

$$C_{\mathrm{c}} = \mu_{\mathrm{c}} C \tag{6a}$$

$$C_{\mathrm{a}} = \mu_{\mathrm{a}} C \tag{6b}$$

$$\bar{C} = C_{\mathrm{a}} + C_{\mathrm{c}} \tag{6c}$$

where μ_{a} and μ_{c} are the dimensionless coefficients depending on the Debye length and pore shape; $\bar{C}$ is the total concentration averaged by pore section.

The Debye length is expressed as:

$$\chi = \frac{1}{zF}\sqrt{\frac{\varepsilon_0 \varepsilon RT}{2C}} \quad (\text{in m}) \tag{7}$$

where ε_0 is the dielectric constant (in F/m) and ε is the dimensionless solution permittivity.

If cation and anion mobilities are equal (that is valid, e.g. for KCl), ion transport depends on average concentrations only:

$$n^{\mathrm{c}} = \frac{C_{\mathrm{c}}}{\bar{C}} = \frac{\mu_{\mathrm{c}}}{\mu_{\mathrm{c}} + \mu_{\mathrm{a}}} \tag{8a}$$

$$n^{\mathrm{a}} = \frac{C_{\mathrm{a}}}{\bar{C}} = \frac{\mu_{\mathrm{a}}}{\mu_{\mathrm{c}} + \mu_{\mathrm{a}}} \tag{8b}$$

The total electrical field E^{t} also consists of the migration component E^{Ω} and diffusion component E^{IP}. Taking into consideration the notation of ion transport number (Eqs. (8a) and (8b)), as well as relationship between the concentration and conductivity in the pore solution:

$$\sigma = (zF)^2 u \bar{C} \quad (\text{in } (\Omega\ \mathrm{m})^{-1}) \tag{9}$$

and using expression (5), E^{IP} can be described as follows:

$$E^{\mathrm{IP}}(\boldsymbol{r},\ t) = -\frac{RT}{F}(n^{\mathrm{c}} - n^{\mathrm{a}})\frac{\mathrm{grad}\bar{C}(\boldsymbol{r},\ t)}{\bar{C}(\boldsymbol{r},\ t)}, \tag{10}$$

Let us assume that the total ion concentration averaged by a pore section consists of the equilibrium concentration $\bar{C}_0$, and the additional concentration, due to the passing electrical current $C^*(\boldsymbol{r}, t)$: $\bar{C}$ $(\boldsymbol{r}, t) = C^*(\boldsymbol{r},\ t) + \bar{C}_0$. Since the process is considered linear (i.e. $C^*(\boldsymbol{r}, t) \ll \bar{C}_0$), Eq. (10) can be reduced to:

$$E^{\mathrm{IP}}(\boldsymbol{r},\ t) \approx -\frac{RT}{F}(n^{\mathrm{c}} - n^{\mathrm{a}})\frac{\mathrm{grad}C^*(\boldsymbol{r},\ t)}{\bar{C}_0} \tag{11}$$

Expression (11) is integrated over the corresponding lengths l_1 and l_2 to obtain the electrical potential

drop at the system of one large and one narrow capillary (Fig. 2b):

$$\Delta U_1^{\mathrm{IP}}(t) = -2\frac{RT}{F}(n_1^{\mathrm{c}} - n_1^{\mathrm{a}})C_{12}^*(t)/\bar{C}_0 \tag{12a}$$

$$\Delta U_2^{\mathrm{IP}}(t) = 2\frac{RT}{F}(n_2^{\mathrm{c}} - n_2^{\mathrm{a}})C_{12}^*(t)/\bar{C}_0 \tag{12b}$$

where $C_{12}^*(t)$ is the positive additional concentration at the large/narrow pore boundary, resulting from the passing electrical current.

In Eqs. (12a) and (12b), we ignored the difference in average concentration in large and narrow pores, which appears to be not of importance for sand.

The IP voltage drop at the system of the large and narrow pores is the sum of the potential differences in Eqs. (12a) and (12b):

$$\Delta U^{\mathrm{IP}} = 4\frac{RT}{F}\Delta n\frac{C_{12}^*(t)}{\bar{C}_0} \tag{13}$$

The migration component of the voltage drop at the system of the large and narrow pores is expressed using Ohm's law:

$$\Delta U^{\Omega} = \frac{j}{\sigma_0}\left(\frac{l_1}{S_1} + \frac{l_2}{\alpha_2 S_2}\right) \tag{14}$$

where σ_0 (in $(\Omega\cdot\mathrm{m})^{-1}$) is the conductivity of equilibrium bulk solution.

Combining Eqs. (13) and (14), one can obtain an expression for polarizability where time dependence is represented by $C_{12}^*(t)$:

$$\eta = 4\frac{RT\sigma_0\Delta n}{jF\bar{C}\left(\frac{1}{L_1}+\frac{1}{\alpha_2 L_2}\right)}C_{12}^*(t) \approx 4 \times \frac{RT\Lambda\Delta n}{jF\left(\frac{1}{L_1}+\frac{1}{\alpha_2 L_2}\right)}C_{12}^*(t) \tag{15}$$

where Λ (in $\mathrm{m}^2\ \Omega^{-1}\ \mathrm{mol}^{-1}$) is the molar conductivity of the solution, $\Lambda=(zF)^2u$; $L_1=S_1/l_1$, $L_2=S_2/l_2$ (in m) are geometric coefficients of the pores.

The IP voltage is usually normalised by the total measured voltage (which includes both Ohm and IP components). However, in ion-conductive rocks and soils, the IP voltage does not exceed few percents of the total voltage. Therefore, normalisation by the migration (Ω) voltage does not produce an important error.

Expression (15) shows that the TD characteristic of the IP is completely defined by the time variation in the ion concentration at the large/narrow capillaries boundary during the polarization cycle (on-time plus off-time). To evaluate the time variation in concentration $C_{12}^*(t)$, we have to solve the equation of diffusion for the conditions defined by the current waveform, as well as by the length and section of large and narrow capillaries.

According to definition of TD transient characteristic (Wait, 1982), let us consider the polarization of a medium caused by the infinite electrical current step. Based on the schematic structure of the current passages in the sand (Fig. 1a), let us also assume that the length of narrow capillaries is significantly less than the length of large ones:

$$l_2 \ll l_1 \tag{16}$$

Assumption (16) produces the short narrow pores (SNP) model. The assumption is not valid for shale sand (Fig. 1c), but appears to be correct for saturated sand (Fig. 1a). The above assumption enables us to simplify the solution of equation of diffusion. Let us further assume that the ion diffusion coefficients in the large and narrow capillaries have the same value.

As long as concentration C^* is averaged by pore section, the equation of diffusion to be solved contains one space and one time variables:

$$D\frac{\partial^2 C^*}{\partial x^2} - \frac{\partial C^*}{\partial t} = q(l_2/2) - q(-l_2/2) \tag{17}$$

where $q(l_2/2)$ and $-q(-l_2/2)$ are the source and discharge active during the polarization pulse, situated on the boundary of narrow/large pores with coordinates $(l_2/2)$ and $(-l_2/2)$, respectively. The capacity of the sources $q=\pm\,\mathrm{d}m/\mathrm{d}t$ (see expression (4)). The sought concentration C^* is obtained by solving Eq. (17):

$$C_{12}^* = \frac{ql_2}{D(S_1+S_2)}\gamma(t,\ \tau) \tag{18}$$

$$\gamma(t,\ \tau) = \frac{1}{\sqrt{\pi}}\sqrt{\frac{t}{\tau}}\left(1-\exp\left(-\frac{\tau}{t}\right)\right) + erfc\left(\sqrt{\frac{\tau}{t}}\right) \tag{19}$$

$$\tau = \frac{l_2^2}{4D} \tag{20}$$

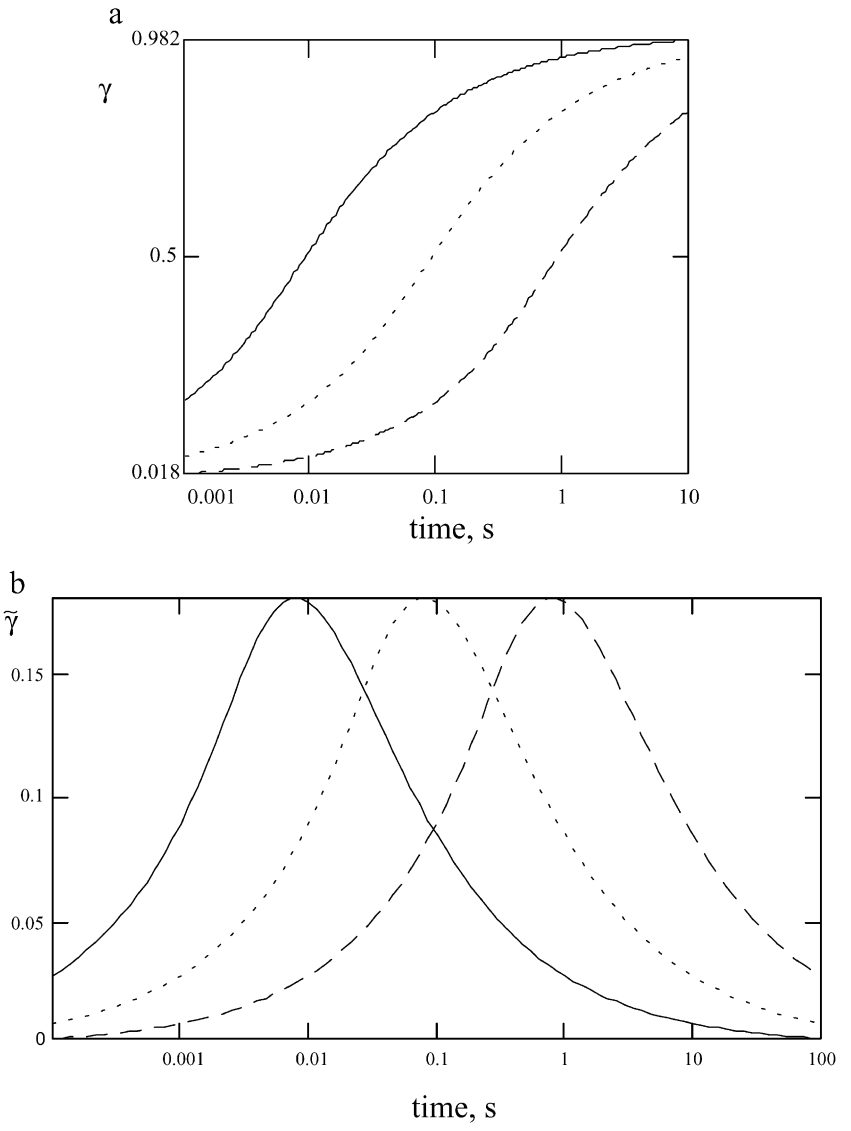


Fig. 3. Normalised transient characteristic of IP (a) and normalised differential polarizability (b) vs. time calculated using Eqs. (19) and (24), respectively. Solid line: $\tau = 0.01$ s, dashed line: $\tau = 0.1$ s, dash-dotted line: $\tau = 1.0$ s. Note that the maxima are slightly shifted to the left from the corresponding values of τ.

where $\gamma(t, \tau)$ is the dimensionless normalised transient characteristic that vary between 0 and 1; τ (in s) is the time constant.

For the solution details, see Appendix A.

Substituting Eq. (4) to Eq. (18) and Eq. (18) to Eq. (15), taking into consideration relationships between ion mobility and diffusion coefficient, and between molar conductivity and ion mobility, one can obtain transient characteristic of polarization:

$$\eta = \frac{4l_2(\Delta n)^2}{(S_1 + S_2)\left(\frac{1}{L_1} + \frac{1}{\alpha_2 L_2}\right)}\gamma(t,\ \tau) = K(\Delta n)^2\gamma(t,\ \tau) \tag{21}$$

where K is the dimensionless geometric factor. Since $\gamma(t, \tau)$ is the normalised transient characteristic, the product $K(\Delta n)^2$ corresponds to chargeability (one can compare with Eq. (2)):

$$\eta_0 = K(\Delta n)^2 \tag{22}$$

Fig. 3a depicts normalised transient characteristics of IP calculated for three values of τ. The differential polarizability and normalised differential polarizability formulae are obtained by differentiating Eqs. (21) and (19) by time logarithm (for details of these parameters, see Section 3):

$$\eta_d = K(\Delta n)^2 \tilde{\gamma} \tag{23}$$

$$\tilde{\gamma} = \frac{\partial \gamma}{\partial(\ln t)} = \sqrt{\frac{t}{4\pi\tau}}\left(1 - \exp\left(-\frac{\tau}{t}\right)\right) = \sqrt{\frac{x}{4\pi}}\left(1 - \exp\left(-\frac{1}{x}\right)\right) \tag{24}$$

where dimensionless $x=\{t/\tau\}$.

The Eq. (24) was used to calculate the normalised differential polarizability shown in Fig. 3b. The abscises of the maxima (t_{max}) in the graphs are related to the values of τ as:

$$t_{max} = 0.794\tau \tag{25}$$

We also show comparison between Cole–Cole and SNP models in TD (Appendix B) and FD (Appendix C).

Eqs. (23) and (24), taking into consideration the notation (20) and formula (25), can be used to interpret experimental data.

3. Experimental

3.1. Studied characteristics

Wait (1982) defined the TD transient characteristic as a time variation in the IP voltage in the course of polarization caused by an infinite unit current step. Komarov (1980) proposed to use the derivative of this transient characteristic by the logarithm of time (differential polarizability, η_d) for theoretical and experimental evaluation of the IP decay. Note that the time derivative of the TD transient characteristic corresponds to IP decay from the pulse of infinitesimal duration. Therefore, η_d represents the product of the above IP decay and the time. In the Cole–Cole model, this differential polarizability versus time produces a maximum corresponding to the maximum in phase shift (φ) versus frequency. The behaviours of the $\eta_d(t)$ and $\varphi(1/\omega)$ are similar.

Komarov (1980) also developed a calculation technique to obtain the $\eta_d(t)$. An electrical current pulse is considered as a superposition of two current steps of the opposite polarity, which correspond to the front and end of the pulse, respectively. Therefore, the signal measured in the off-time represents the finite difference of the two transient characteristics, mutually displaced along the time axis for the duration of the pulse. If the transient characteristic versus time logarithm is nearly linear, the differential polarizability can be obtained by dividing the measured signal by the corresponding coefficient:

$$P = \log(t_i + t) - \log(t) = \log(1 + t_i/t) \tag{26}$$

where t is the measurement time, counted from the end of the pulse, t_i is the duration of the pulse.

This differential polarizability presents the following advantages comparing to IP decay:

- the η_d slowly changes with time (for data corresponding to the constant phase angle model) or contains maximum (for data corresponding to the Cole–Cole and similar models), which makes it easier to analyse instead of IP decay;
- the data obtained using various duration of pulses can be rewritten in term of differential polarizability and then compared;
- the behaviour of $\eta_d(t)$ is similar to $\phi(1/\omega)$, which allows easy comparison of data obtained in TD and FD.

Examples of the use of differential polarizability are abundant in Russian publications. One can find examples presented in English in Mikhailov et al. (1986).

3.2. Experimental procedure

The measurement setup is shown in Fig. 4. STROB-M (Shereshevsky et al., 1999) is a multifunctional measuring device developed in the Laboratory of Electrical and Electromagnetic Methods of the Russian Institute of Exploration Geophysics (VIRG-Rudgeofizika). It was designed for field and laboratory measurements in the TD electromagnetic and IP methods. The dynamic range of the measured voltage values is 160 dB, time range varies from 5 μs to 10 min. Measurements are controlled by a built-in microprocessor.

The laboratory transmitter used in the experiments was capable of switching the current off within several microseconds. The receiver and transmitter were synchronised using quartz timer. Rectangular current pulses of opposite polarity were transmitted through the sand samples. We used pulses and pauses of the same duration. The measurements were conducted during off-time, when the transmitter resistivity was 200 MΩ, input resistivity of the measuring device was 90 MΩ. In the above conditions, the influence of the sample discharge through the current and potential circuits was minimal. Voltage was measured in the pauses between the current pulses of the

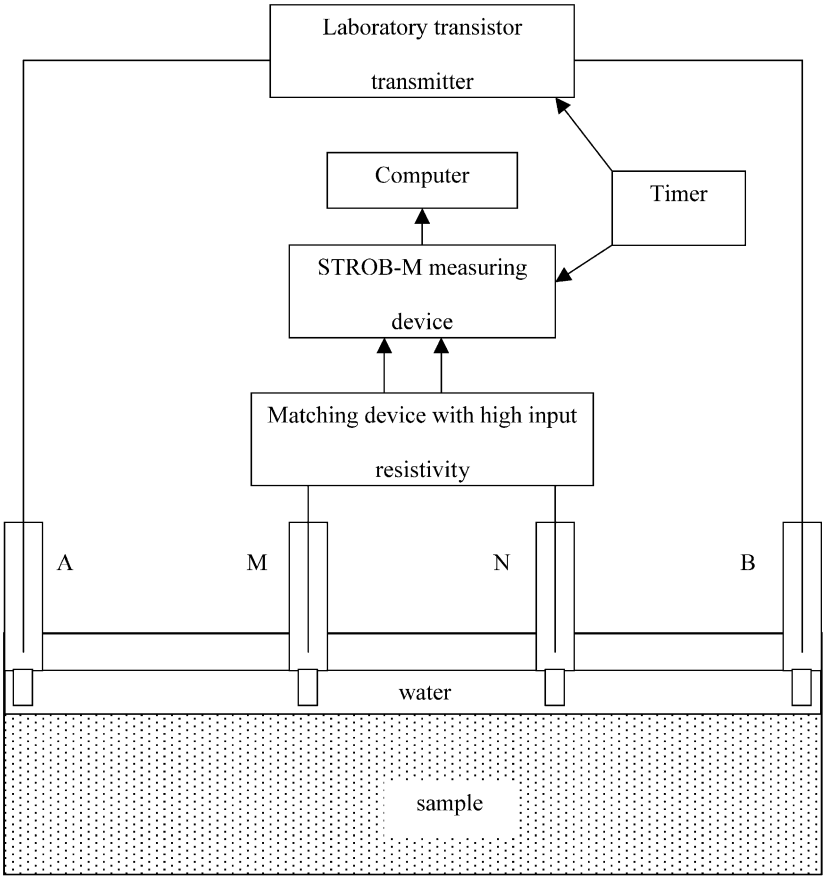


Fig. 4. Block-diagram of the laboratory TD spectral IP measurement system. Cu–CuSO4 nonpolarizing electrodes (A, M, N and B) were positioned in the water layer (2–3 mm) that covered the sand that filled the vessel. A symmetric measurement array was used.

opposite polarity. As usual in the TD technique, the voltage decay was sampled through a sequence of windows. We sampled each IP decay 50–200 times. The value recorded for each window is expressed as:

$$\eta = \frac{100}{U_{\mathrm{i}} \cdot (t_{k+1} - t_k)} \int_{t_k}^{t_{k+1}} U(t)\mathrm{d}t \quad (\text{in } \%) \tag{27}$$

where t_k, t_{k+1} are the limits on the integration and U_i is the time average of the measured voltage during the current on-time.

To reduce electromagnetic noise, the pulse duration, time windows and number of stacked pulses were varied when early (0.001–0.1 s), medium (0.1–10 s) or late (10–300 s) polarization was studied. The pulses of 0.1–300 s were used depending on the time range studied. When the early time polarization was studied, the short pulses (0.1–1 s) and narrow time windows (up to 10 ms) were used; up to 100 pulses was stacked. When the late time polarization was studied, the long pulses (30–300 s) were used, time windows became longer (up to 160 ms); up to 10 pulses were stacked.

As long as the stack mode was used, the simple method, described in Section 3.1, was modified to obtain the differential polarizability. The measured voltage decay can be represented by the superposition of transient characteristics corresponding to each front and end of the pulses:

$$\frac{\Delta U_{\mathrm{p}}(t)}{\Delta U_{\mathrm{i}}} = \eta_0 \sum_{k=0}^{N} (-1)^k (\Psi(t + (2k+1)t_i) - \Psi(t + 2kt_i)) \tag{28}$$

where N is the number of stacked pulses, the product $\eta_0\Psi(t)$ is an experimental transient characteristic, t is the time counted from the end of the last pulse. Unknown quantity $\eta_0\Psi(t)$ is represented as a polynomial:

$$\eta_0 \Psi(t) = \sum_{m=1}^{M} a_m (\log(t))^m \tag{29}$$

of order $M=3$. Coefficients of polynomial am are obtained using the least-square method. The differential polarizability was calculated using logarithmic differentiation of the determined polynomial. We applied the above algorithm to the IP decay in a floating time window to obtain the better correspondence between the measured and approximate voltages.

First measurements on the tap water (salinity 100 mg/l) without sand were conducted to control accuracy of the measurement setup. Fig. 5 shows the apparent differential polarizability obtained in the water. The values stipulated by the electrode polarization and other types of noise greatly diminished with time, and after the first 20 ms read less than 0.01%.

Then two samples of natural silicate sand with different grain-size distribution were studied. The coarse-grained white sand No. 1 was taken from the beach of Gulf of Finland. The sample No. 2, from eluvium, contained dust particles and had reddish tint, showing presence of colloidal iron-containing minerals. The unwashed samples were saturated with the same tap water.

The natural silicate sand of mixed composition was sieved to four grain sizes: 0.8–1.5, 0.5–0.8, 0.3–0.5 and 0.2–0.32 mm. Clay and dust particles were removed by repetitive washing of the sand in the same tap water. The IP decays of these sieved sands were measured.

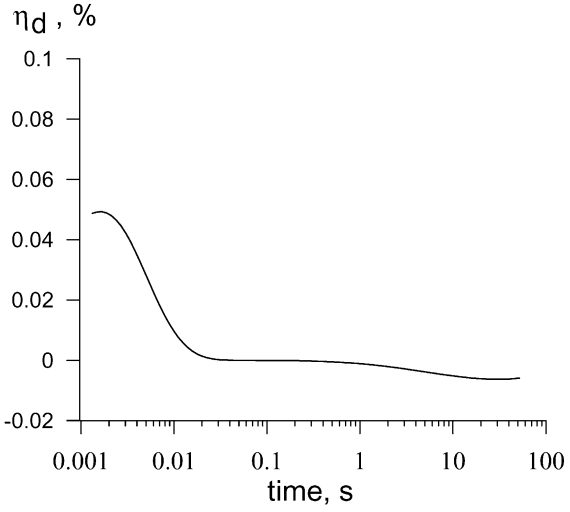


Fig. 5. Apparent differential polarizability obtained on the water. Note that when $t > 0.02$ s, the apparent differential polarizability becomes lower than 0.01%. Low values stipulated by the electrode polarization confirm accuracy of the measurement setup.

3.3. Results

3.3.1. Natural sands

The experimental data are presented in the Fig. 6 as the differential polarizability versus time. The differential polarizability values shown in the graphs are noticeably low (less than 1%), which is typical of sands. The η_d decreases with time. Sands studied in FD show the same behaviour in phase angle $\phi(1/\omega)$ (Vanhala, 1997). Therefore, neither the Cole–Cole nor constant phase angle model corresponds with the data in the entire time range.

The grain-size distributions show that for the first sample the dominant fraction is two steps greater than that for the second sample. Small particles are less abundant in the first sample.

One can see the following features of differential polarizability in these natural samples:

- in the early time range (0.001–0.1 s), the first sample shows greater values than the second sample;
- in the medium time range (0.1–1 s), the behaviour of the both samples is similar;
- in the end of medium and the late time range (>1 s), the differential polarizability is small but well readable for the first sample, and practically disappears for the second sample.

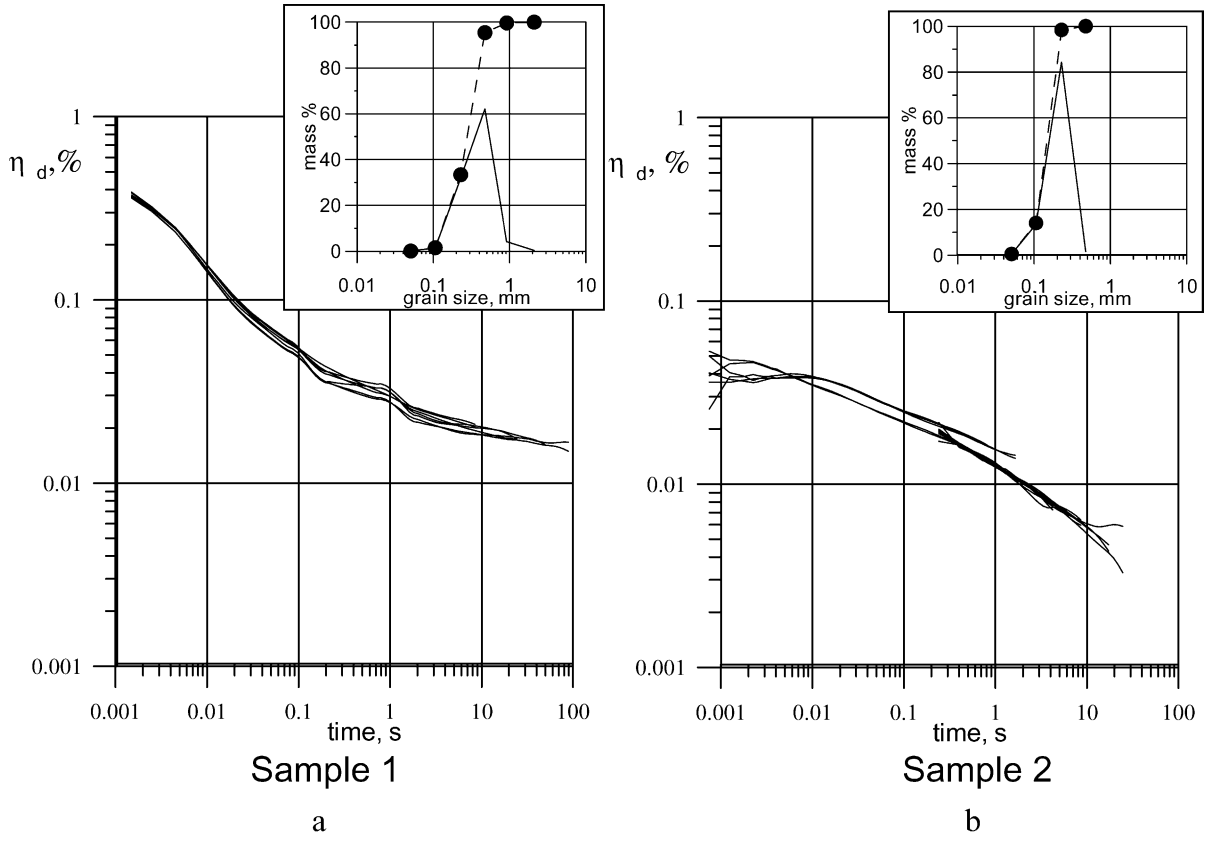


Fig. 6. Experimental data on differential polarizability in the water-saturated natural sand (a—sample 1, b—sample 2). Each continuous curve represents the differential polarizability obtained from current pulse of the constant duration t_i. Convergence of data obtained from pulses of various duration confirms accuracy of measurements and data treatment. Each measurement was repeated four to six times. To demonstrate good reproducibility, all the data are presented. Grain size distribution is given in solid lines and the corresponding cumulative in dashed lines with filled circles.

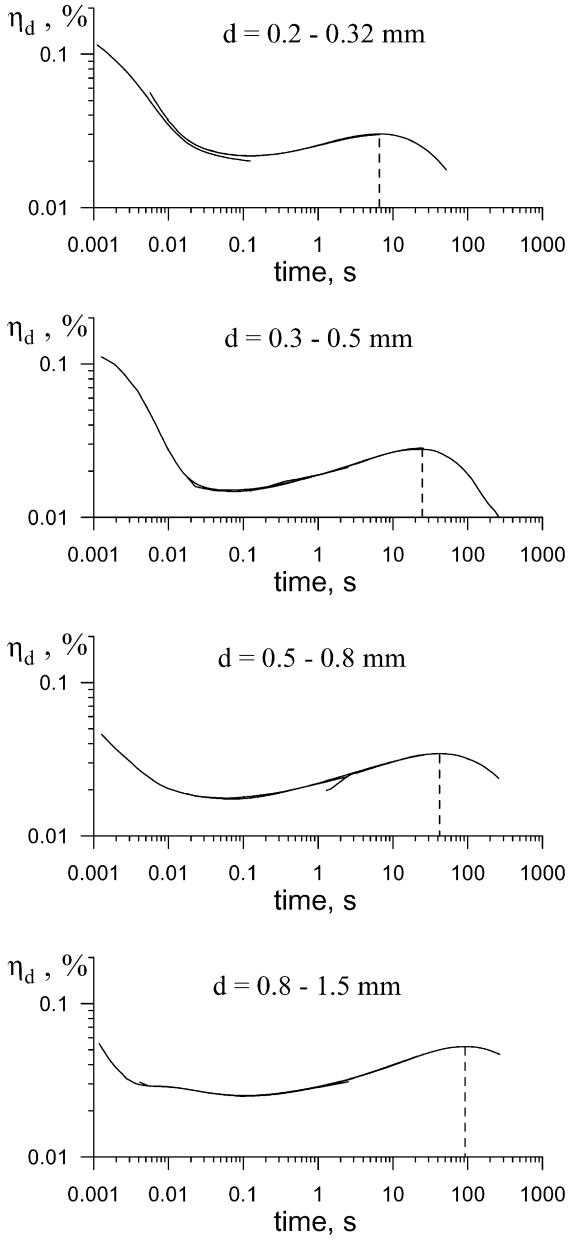


Fig. 7. Experimental data on the differential polarizability in the water-saturated sieved sand. Each continuous curve represents differential polarizability obtained from the current pulse of a constant duration t_i. Note very close convergence of the data obtained from pulses of various duration.

3.3.2. Sieved sands

The data are presented in Fig. 7 as differential polarizability versus time. The sharp decrease in the η_d values can be observed at early times. A similar decrease was also observed in pure water measurements. It is probably due to the polarization processes that take place at the potential and current electrodes. The electrodynamic (Maxwell–Wagner) polarization may also occur in early times range for the sand (see also Fig. 6, first sample) (Dukhin, 1971; Olhoeft, 1985). In all four graphs for sieved sands, a maximum of the differential polarizability (0.025–0.05%) is observed in the 7–100 s range. One can see that the time of maxima increases when average grain size increases. The parts of spectra containing maxima (0.1–300 s) can be approximated by the Cole–Cole formula.

4. Discussion

Like the Cole–Cole and Marshall–Madden models, the proposed SNP model is valid for a single type of polarization sources in soil. Therefore, the model is comparable with the data obtained in sieved sands where the lengths of narrow and large current passages are nearly constant.

Based on the experimental data (Fig. 7), we obtain the values of τ for each grain size of the sand from times of maximum (see Eq. (25)) to verify the correctness of the SNP model. The effective lengths of the narrow pores (l_2 values) in Table 1 are calculated on the basis of Eq. (20) using the diffusion coefficient for potassium chloride in bulk solution ($\sim 10^{-9}$ m^2/s). In a system of spherical grains, the ratio of the particle size and the effective length of the narrow pore (linear size of the grain

Table 1
Experimental characteristics of differential polarizability in the sands of different grain sizes

Sample no.	Grain size (mm)	Average particle diameter (d, mm)	t_{max} (s)	τ (s)	l_2 (mm)	d/l_2
1	0.8–1.5	1.15	90	110	0.66	1.7
2	0.5–0.8	0.65	40	50	0.45	1.4
3	0.3–0.5	0.40	22	28	0.33	1.2
4	0.2–0.32	0.26	7.0	8.8	0.19	1.4

t_{max}—the time of maximum of differential polarizability, τ—the time constant according to Eq. (25), l_2—the length of narrow pore according to Eq. (20).

contact) is nearly constant ($d/l_2 \approx l_1/l_2 \sim$ constant). Close values of the calculated ratio d/l_2 for four sieved sands confirm the accuracy of the proposed model (Table 1). The average value of the above ratio is 1.4. Note that the calculated ratio strongly depends on the diffusion coefficient value.

Based on the aforementioned average value of d/l_2, the time constant τ and the sand grain size (d, mm) can be related as:

$$d \approx 0.09\sqrt{\tau} \tag{30}$$

where the empirical factor 0.09 is given in mm·s$^{-1/2}$ (Fig. 8).

The obtained ratio d/l_2 allows estimation of chargeability η_0 of sands on the basis of the SNP model. Let us assume that ${l_1}^2 \sim S_1$, ${l_2}^2 \sim S_2$. In the above conditions, the geometric factor $K \sim 4 \cdot {l_2}^2/{l_1}^2$ (see Eq. (21)). $K \sim 4 \cdot {l_2}^2/d^2 \sim 4 \cdot 1.4^{-2} \approx 2$ because $l_1 \sim d$. Thus, $\eta_0 \sim 2(\Delta n)^2$, and the range of difference of ion transport number 0.1–0.05 produces $\eta_0 \sim 2\ldots0.5\%$ that corresponds to the experimental data.

The data obtained on the natural sand (Fig. 6) do not contain maxima in the range 1–100 s, where sieved sands show maximum. As long as the different lengths of narrow and large pores occur in natural sand, different values of τ are produced. If the soil presents a monotonous distribution of τ in the studied time range (which corresponds to the monotonous grain-size distribution), the differential polarizability versus time (or phase angle versus frequency) can be monotonous. In case of constant distribution of τ, the constant phase angle occurs. Thus, models containing single τ do not properly describe soils with mixed grain size.

We assume that the quantity of particles of a given size "d" in a volume unit determines the contribution of polarization with the time constant "τ" to the total polarization. Therefore, the inversion of IP spectra to grain-size distribution is theoretically possible.

The frequency dependency of IP contains three parameters in the Marshall–Madden's model: selective zone length (l_2), ratio of selective/nonselective lengths (l_2/l_1) and number of cation transport (n^c). The proposed SNP model differs from the Marshall–Madden model because the l_1 is assumed to be noticeable greater than l_2. Thus, the model contains only two variables (Δn) (which related to n^c) and τ (which related to l_2). Therefore, the SNP model represents a special case of the Marshall–Madden's model proposed for sands. Since the SNP model contains only two variables (one of which represents the multiplier), this model appears to be useful for an inversion of IP spectra to obtain grain-size distribution of sands.

Comparison of the IP spectra of the 1st and 2nd natural sand samples shows that for the coarse-grained sand the differential polarizability in the late time range is noticeable higher than that for the fine-grained sand, which is in agreement with other data obtained for sands (Vanhala, 1997).

In the early time range, large values of η_d were obtained only for washed sieved sands and for unwashed natural sand No. 1. However, the first sample was naturally washed because it was taken from the sea beach. The second sample, which contained dust and colloidal particles, did not show large polarization in the early time range. Although early polarization was observed only in few samples, we can assume that colloidal and dust particles control the early polarisation.

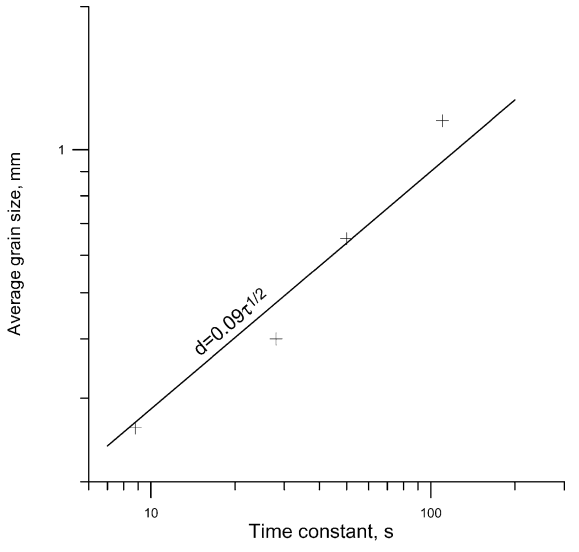


Fig. 8. Average grain size vs. time constant for sieved sands: crosses are experimental data, solid line is the approximation.

5. Conclusions

• Based on the phenomenon of membrane polarization, theoretical characteristics of spectral IP in pure sands, called small narrow pores (SNP) model, are obtained. The proposed model represents a special case of the Marshall–Madden model. The SNP model is based on the assumption that the narrow pores are noticeably shorter than the large ones. This assumption allows us to use fewer model parameters than in the Marshall–Madden and Cole–Cole models. Thus, the validity of the proposed model is limited, yet the model seems to be useful for inversion of IP spectra for obtaining grain-size distribution of sands.

• According to the proposed model, the time constant τ in sieved sands increases with the growth of the grain size as: $\tau \sim d^2$.

• Two samples of natural sands with different grain-size distribution show IP spectra decreasing with time. In the early time range, these spectra are assumed to be related to the Maxwell–Wagner polarization. The spectra in the medium and late time ranges are produced by the membrane polarization, characterised by various relaxation times. This variety of the relaxation time is related to different grain sizes in natural sands, which result in different lengths of large and narrow current passages.

Acknowledgements

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Appendix A. Solution of the equation of diffusion for the SNP model

(1) An instantaneous source of Q (mol) is placed on boundary of two tubes with different sections S_1 and S_2 and the coefficients of diffusion D_1 and D_2 (Fig. 9).

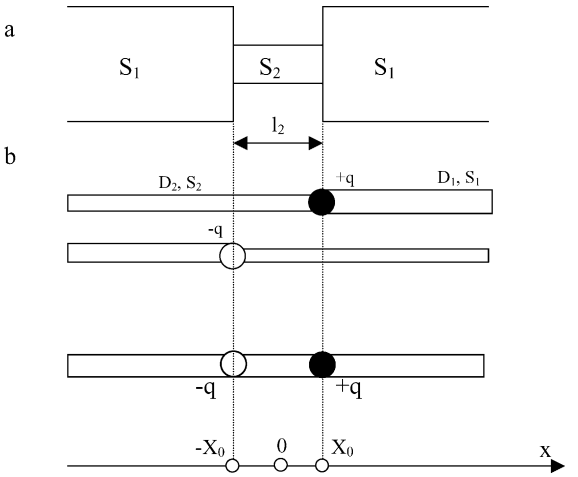


Fig. 9. Contacts of two large pores and a narrow pore (a) and equivalent scheme for solving the equation of diffusion (b). Source $+q$ and discharge $-q$ model concentration surplus and deficiency produced by the polarizing current.

The fundamental solution of equation of diffusion for each medium is expressed as (Carslow and Eger, 1964):

$$C_1 = \frac{a_1}{\sqrt{\pi D_1 t}} \exp\left(-\frac{(x-x_0)^2}{4D_1 t}\right) \tag{A1.1}$$

$$C_2 = \frac{a_2}{\sqrt{\pi D_2 t}} \exp\left(-\frac{(x-x_0)^2}{4D_2 t}\right) \tag{A1.2}$$

where a_1, a_2 are constants depending on the S_1, S_2, D_1, D_2.

To determine a_1 and a_2, two conditions are used:

$$C_1 = C_2(x = x_0) \tag{A1.3}$$

$$S_1 \int_{-\infty}^{x_0} C_1 \mathrm{d}x + S_2 \int_{x_0}^{\infty} C_2 \mathrm{d}x = Q \tag{A1.4}$$

After substitution of C_1, C_2 in Eqs. (A1.3) and (A1.4) and simplifications, the conditions are reformulated:

$$\frac{a_1}{\sqrt{D_1}} = \frac{a_2}{\sqrt{D_2}} \tag{A1.5}$$

$$a_1 S_1 + a_2 S_2 = Q \tag{A1.6}$$

The constants a_1 and a_2 are defined from Eqs. (A1.5) and (A1.6):

$$a_1 = \frac{Q\sqrt{D_1}}{S_1\sqrt{D_1} + S_2\sqrt{D_2}} \tag{A1.7}$$

$$a_2 = \frac{Q\sqrt{D_2}}{S_1\sqrt{D_1} + S_2\sqrt{D_2}} \tag{A1.8}$$

If $D_1 = D_2$ and $a_1 = a_2$, the concentration in the both media is expressed as:

$$C = \frac{Q}{(S_1 + S_2)\sqrt{\pi D t}} \exp\left(-\frac{(x - x_0)^2}{4Dt}\right) \tag{A1.9}$$

(2) The negative source introduced at the point $-x_0$ (Fig. 9) is taken into consideration on the basis of superposition. Thus, the concentration caused by positive and negative sources at the points $+x_0$ and $-x_0$ is expressed as:

$$C = \frac{Q}{(S_1 + S_2)\sqrt{\pi D t}} \times \left[\exp\left(-\frac{(x - x_0)^2}{4Dt}\right) - \exp\left(-\frac{(x + x_0)^2}{4Dt}\right)\right] \tag{A1.10}$$

where $2x_0 = l_2$. Note that $C|_{x=0} \equiv 0$ at any time.

The concentration at the point x_0 is expressed as:

$$C = \frac{Q}{(S_1 + S_2)\sqrt{\pi D t}} \left[1 - \exp\left(-\frac{l_2^2}{4Dt}\right)\right] \tag{A1.11}$$

(3) During an infinite current step, the sources of capacity $\pm q$ (mol/s) produce concentration $C_{12}^*(t)$ obtained by integration of Eq. (A1.11):

$$C_{12}^*(t) = \frac{Q}{(S_1 + S_2)\sqrt{\pi D}} \int_0^t \times \left[1 - \exp\left(-\frac{l_2^2}{4D(t - t')}\right)\right] \frac{\mathrm{d}t'}{\sqrt{t - t'}} \tag{A1.12}$$

The result of integration of Eq. (A1.12) produces the solution of equation of diffusion representing time dependence of transient characteristic:

$$C_{12}^* = \frac{ql}{D(S_1 + S_2)} \times \left(\frac{1}{\sqrt{\pi}}\sqrt{\frac{t}{\tau}}\left(1 - \exp\left(-\frac{\tau}{t}\right)\right) + erfc\left(\sqrt{\frac{\tau}{t}}\right)\right) \tag{A1.13}$$

where q is the modulus of intensity of positive and negative sources in mol/s,

$$\tau = \frac{l_2^2}{4D} \tag{A1.14}$$

Since τ is measured in time units, it can be considered as the "relaxation constant" of the polarization process.

Appendix B. Correspondence between Cole–Cole and SNP models in time domain

The logarithmic derivative Y of function Ψ_c (see Eq. (2)) is expressed as:

$$Y(x) = \frac{\mathrm{d}\Psi_c(x)}{\mathrm{d}\ln(x)} = -\sum_{k=1}^{\infty}(-1)^k \frac{ck(x)^{ck}}{\Gamma(1 + ck)} \tag{A2.1}$$

This normalised differential polarizability $Y(x)$ contains a maximum (Fig. 10). The maximum is shifted from the value of "1" at the x-axis. The value of t_{max} relates to τ as:

$$t_{max} = N(c)\tau \tag{A2.2}$$

where $N(c)$ is the dimensionless shift correction coefficient depending on the exponent c in Eqs. (1) and (2). The values of $N(c)$ and corresponding maxima of Y are shown in Table 2.

We found that the value of Cole–Cole exponent $c = 0.62$ produces close maxima of $Y(x)$ and $\gamma(x)$, calculated on the basis of Eqs. (A2.1) and (24).

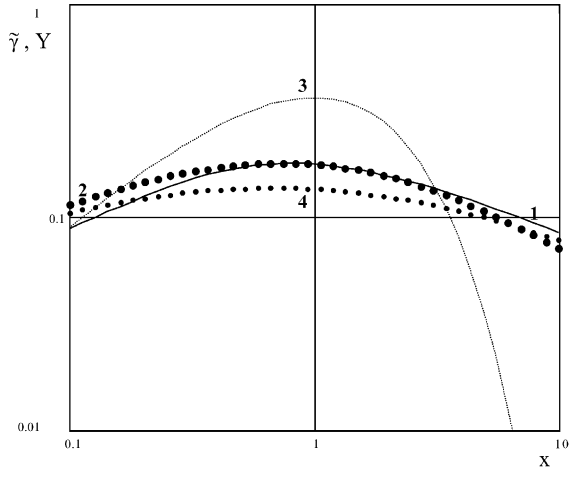


Fig. 10. Comparison between the Cole–Cole and SNP models in TD. Normalised differential polarizability for SNP ($\tilde{\gamma}(x)$) is calculated using Eq. (24) (graph 1). Normalised differential polarizability for the Cole–Cole model ($Y(x)$) is calculated using Eq. (A2.1) with various exponents [$c=0.62$ (graph 2), $c=1$ (graph 3), $c=0.5$ (graph 4)]. Dimensionless argument $x=t/\tau$. Note that the maximum of $Y(x)$ is not shifted from the value of "1" at x-axis if the Cole–Cole exponent $c=1$ only. The smaller is the c, the larger is the shift between t_{max} and τ.

One can compare the corresponding $N(c)$ equal to 0.735 (see Table 2) with the multiplier determining shift between maximum of normalised differential polarizability and τ according the SNP model (see Eq. (25)). The difference between abscissas of maxima of functions Y and γ is 7.4%. Thus, these functions are in satisfactory correspondence, and in practice one can use any of them. For comparison, we also show graphs of $Y(x)$ corresponding to $c=1$ and 0.5 (Fig. 10).

Table 2
Parameters of the Cole–Cole model in the time domain

c	N(c)	Y_{max}
0.1	0.566	0.0251
0.2	0.580	0.0508
0.3	0.602	0.0778
0.4	0.634	0.1067
0.45	0.653	0.1222
0.5	0.675	0.1384
0.55	0.698	0.1554
0.6	0.724	0.1734
0.62	0.735	0.1810
0.63	0.741	0.1848
0.65	0.752	0.1926
0.7	0.782	0.2129
0.75	0.814	0.2346
0.8	0.848	0.2577
0.85	0.883	0.2825
0.9	0.921	0.3090
0.95	0.960	0.3374
1.0	1.000	0.3679

c—the Cole–Cole exponent, $N(c)$—the shift correction coefficient, Y_{max}—the maxima of the normalised differential polarizability.

Appendix C. Correspondence between the Cole–Cole and SNP models in frequency domain

Applying the Laplace–Carson transform to the SNP normalised transient characteristic (Eq. (19)), we can reformulate the SNP model in the FD as follows:

$$\rho(\omega)=\rho_0\left(1-\eta_0\left(1-\frac{1-\exp(-2(i\omega\tau)^{1/2})}{2(i\omega\tau)^{1/2}}\right)\right) \tag{A3.1}$$

where $\rho(\omega)$ is the frequency dependent resistivity (in Ω·m); ρ_0 is the DC resistivity (in Ω·m); η_0 is the chargeability (in V/V) given in Eq. (22).

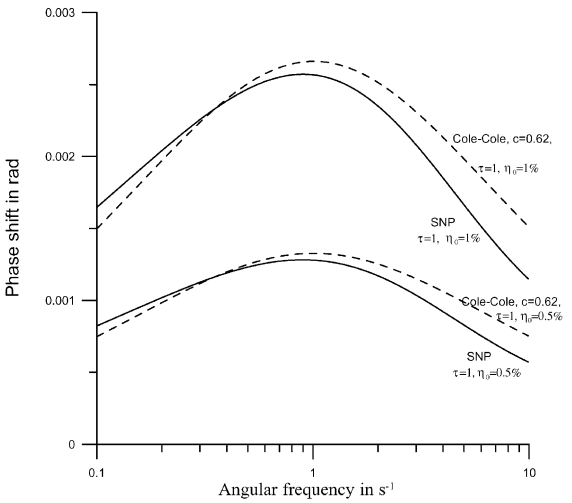


Fig. 11. Comparison between the Cole–Cole and SNP models in FD. Cole–Cole graphs are dashed line, SNP graphs are continuous lines. The graphs of phase shift are calculated using two values of chargeability typical of clean sands (1% and 0.5%). The value of Cole–Cole exponent c, which gives correspondence to the SNP model, was determined in the previous section ($c=0.62$).

One can compare the phase spectra produced by the SNP and Cole–Cole models. Based on Eq. (A3.1), we calculated the phase shift of $\rho(\omega)$ as:

$$\varphi = a\tan\left(\frac{Im(\rho(\omega))}{Re(\rho(\omega))}\right) \quad \text{(A3.2)}$$

The Cole–Cole phase shift was calculated on the basis of Eq. (1). Fig. 11 shows the Cole–Cole and SNP phase shifts calculated using values of chargeability typical of clean sands (1% and 0.5%), and the Cole–Cole exponent determined in Appendix B ($c = 0.62$). One can note the agreement between Cole–Cole and SNP phase spectra.

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